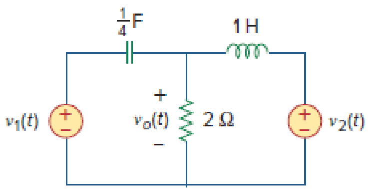
Chapter 16, Problem 88P

Problem

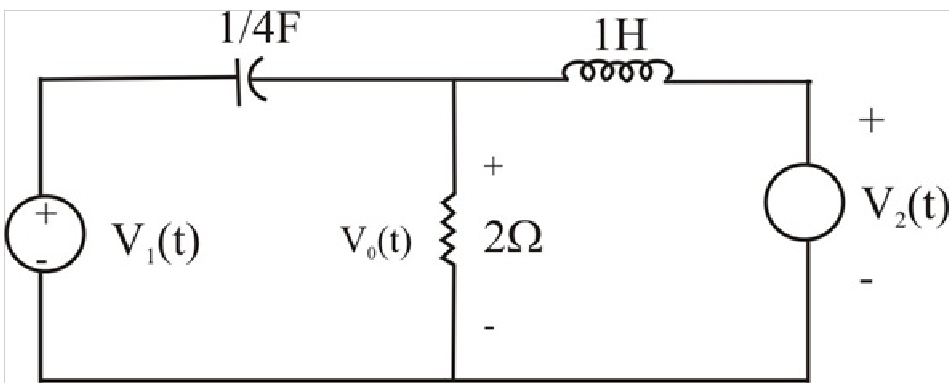
Develop the state equations for the circuit shown in Fig. 16.102.

Figure 16.102:



Step-by-step solution

Step 1 of 2



Step 2 of 2

First select the Inductor current I_l' current flowing left to R_1GHT and capacitor voltage V_C (Voltage positive on left and negative on right) to be state variables.

Applying KCL we get;

$$-\frac{V_C'}{4}+\frac{V_O}{2}+I_L'=0; V_C'=4I_L'+2V_O$$

$$I_L'=V_O-V_2; V_O=-V_C+V_L$$

$$V_C'=4I_L'-2V_C+2V_1; I_L'=-V_C+V_1-V_2$$

$$\begin{bmatrix} I_L' \\ V_L' \end{bmatrix}=\begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} I_L' \\ V_C \end{bmatrix}+\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} V_1(t) \\ V_2(t) \end{bmatrix}$$

$$V_O(t)=\begin{bmatrix} 0 & -1 \end{bmatrix}\begin{bmatrix} I_L' \\ V_C \end{bmatrix}+\begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} V_1(t) \\ V_2(t) \end{bmatrix}$$